



COVID-19 Scenario Modeling Hub

COVID-19 Scenario Modeling Hub Report

Round 20

16 September, 2026, Scenario Modeling Hub Team¹

Overview

The 20th round of COVID-19 projections focuses on the impact of the frequency of COVID-19 boosters for high-risk groups and vaccination coverage. With this round, SMH aims to provide insight into the impact of continued recommendations that high risk populations receive a booster twice per year, one prior to the summer wave of COVID-19, and one prior to the winter wave. This round also aims to assess the impact of current vaccination coverage as compared to either no vaccination or increased coverage levels. The projection period for Round 20 spans two years, from Sunday, June 8, 2025, to Saturday, June 5, 2027, including one retrospective year and one prospective year. Detailed scenario descriptions and setting assumptions are provided in the [COVID-19 Scenario Modeling Hub GitHub repository](#).

Timing and overall dynamics:

- COVID-19 is expected to follow the same seasonal patterns of activity as in recent years, with two waves peaking in September 2026 and January 2027 nationally. Based on the 50% projection interval, and in all vaccination scenarios, hospitalizations and deaths are expected to continue to decline compared to last year.
- In the business as usual vaccine scenario, we project 240,000 cumulative hospitalization (95%PI 183,000 - 330,000) between June 2026-June 2027 and 11,500 deaths (95% PI 8,000 - 16,300) due to COVID-19, nearly equally spread out between the summer and winter waves.

Vaccine impact:

- All vaccination strategies are projected to significantly reduce disease burden compared to no vaccination, irrespective of assumptions about the coverage and frequency of the vaccination campaign. A business as usual vaccine coverage mirroring last year's uptake would reduce hospitalizations by 16% (95% CI 8% - 24%) and deaths by 20% (95% CI 10% - 30%), compared to no vaccination. This represents 53,000 (95% CI 21,000 - 84,000) hospitalizations and 3,700 (95% CI 840 - 6,500) deaths averted. An additional spring campaign targeting half of the high-risk individuals who receive vaccination in the fall would avert an extra 4% (95% CI 2% - 5%) of hospitalizations and 6% (95% CI 3% - 8%) deaths, corresponding to 9,400 (95%CI 6,100 - 13,000) hospitalizations and 730 (95%CI 170 - 1,300) deaths averted.
- Greater disease burden reductions would be obtained with a higher COVID-19 booster coverage that mirrors the uptake of the flu vaccine. An annual flu-like campaign would avert 31% (95% CI 17% - 46%) of hospitalizations and 36% (95% CI 19% - 52%) of deaths, corresponding to 100,000 (95%CI 41,000 - 160,000) hospitalizations and 6,300 (95%CI 2,100 - 10,000) deaths averted. An additional spring campaign targeting half of the high-risk individuals who receive vaccination in the fall would avert an extra 8% (95% CI 4% - 12%) of hospitalizations and 12% (95% CI 7% - 17%) deaths, corresponding to 15,000 (95%CI 9,600 - 20,000) hospitalizations and 1,100 (95%CI 390 - 1,900) deaths averted.

A few additional notes:

- All models project a semi-annual pattern of activity, with consistent timing. In one model, high vaccine coverage scenarios D and E eliminate the winter 2027 peak.
- The retrospective period allowed teams to calibrate their models to recent disease severity and seasonal patterns and to evaluate a counterfactual Spring 2026 vaccination campaign in scenarios C and E. Estimates of cumulative burden and burden averted by vaccination are provided for the prospective projection period, June 2026–June 2027
- We assumed VE of COVID-19 boosters would be 55% against hospitalization at the start of the fall 2026 vaccination campaign, based on recent US data. Both effectiveness of boosters against existing and new variants and rate of immunity waning after multiple booster shots and repeat infections remain unclear. Further, we assumed that the

¹Compiled by Lucie Contamin, Shaun Truelove, Cécile Viboud.

vaccine administered during the spring 2026 campaign in scenarios C and E would have the same formulation as that of the 2025 fall vaccine, so that effective VE would be expected to be lower due to immune escape. The VE of the spring vaccine was left at the discretion of the modeling teams.

- The pace of immune escape was left at teams' discretion. Teams assumed continuous immune escape rather than discrete variants, mirroring observations of evolutionary changes in the last year. We did not consider the impact of a significant new variant that would have accumulated a large amount of antigenic changes over a very short period, nor increase in transmissibility (akin to Delta or Omicron variants). We also assumed that the intrinsic severity (i.e., severity in naive populations) of future circulating strains would remain similar to that of the Omicron lineages.
- There is sizable heterogeneity between states and between individual models in seasonality and vaccine effects. In particular, a more pronounced peak of summer COVID-19 activity is projected in Southeastern and middle Atlantic states.

Round 20 Specifications

	No vaccine recommendation	Vaccine recommended to all ages once a year in the fall • Vaccination starts Aug 13, 2025 and Aug 14, 2026	Vaccine recommended to all ages; twice a year for high-risk • Fall first-dose vaccination starts Aug 13, 2025 and Aug 14, 2026 • Second dose vaccination for high-risk starts February 15, 2026 • Coverage of second dose is 50% of that of first dose
No vaccine recommendation • No new vaccination during the 2026-27 season	Scenario A		
Business as usual coverage • Uptake in all population subgroups, including high-risk, is the same as reported for COVID-19 in 2025-26		Scenario B	Scenario C
Optimistic vaccination coverage • Uptake in all population subgroups saturates at levels of the 2025-26 influenza vaccine		Scenario D	Scenario E

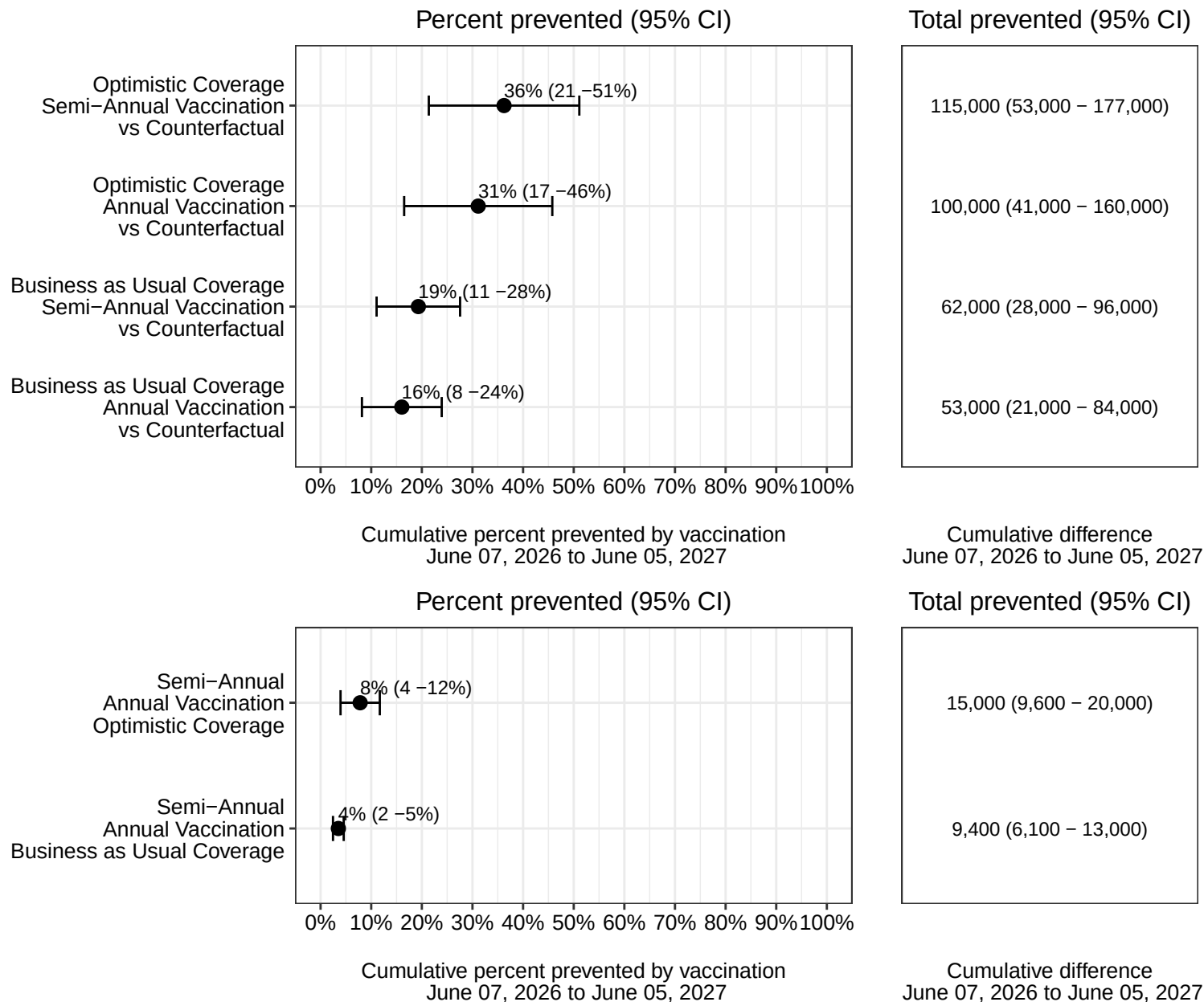
National Scenario Comparative Impacts

Estimates (cumulative pooled differences) of relative impact and cumulative hospitalizations averted in the US, overall population from June 07, 2026 to June 05, 2027.

Impacts of COVID-19 Immunization Scenarios, National

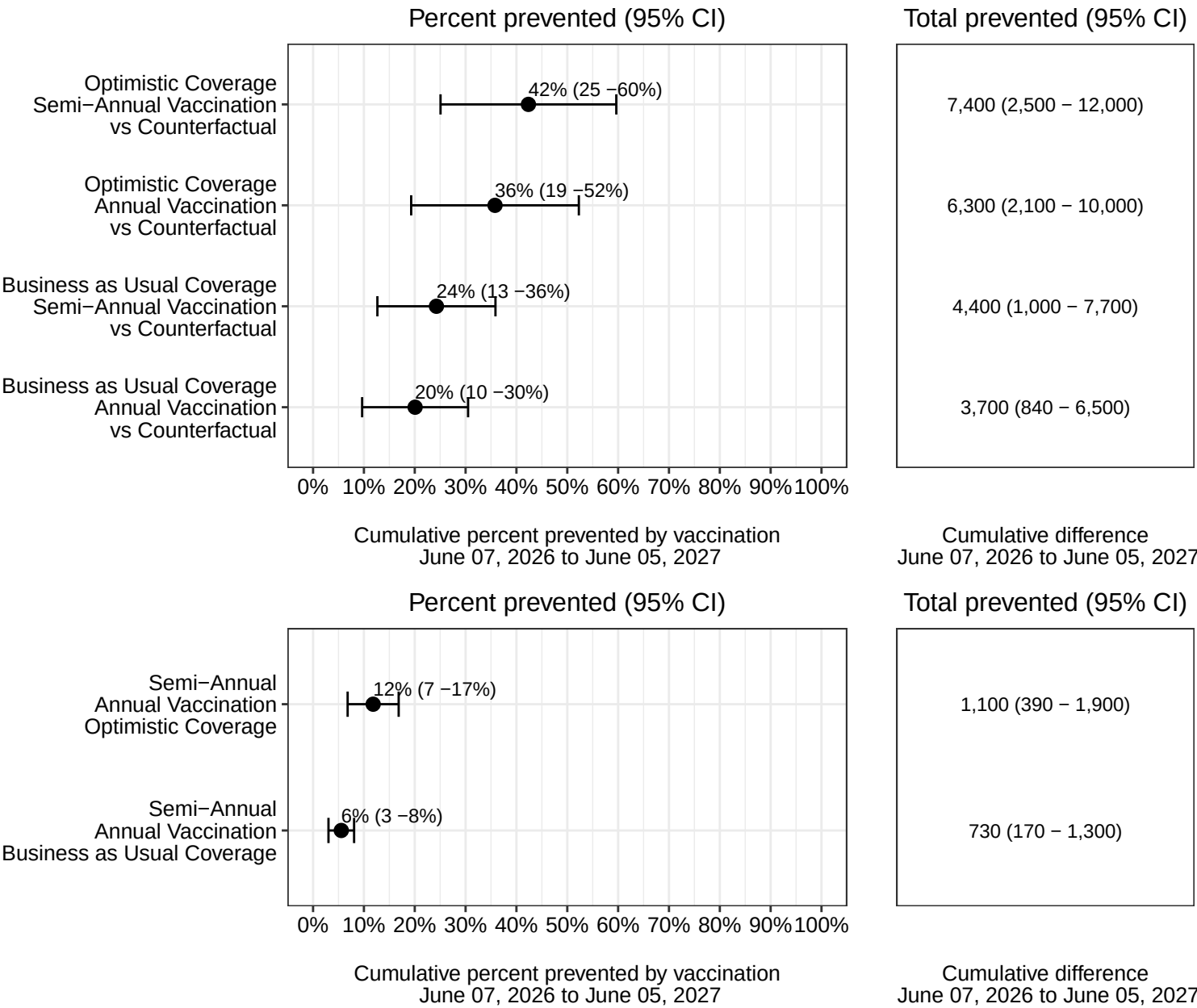
Hospitalization

Percent and number of hospitalizations averted by immunization



Death

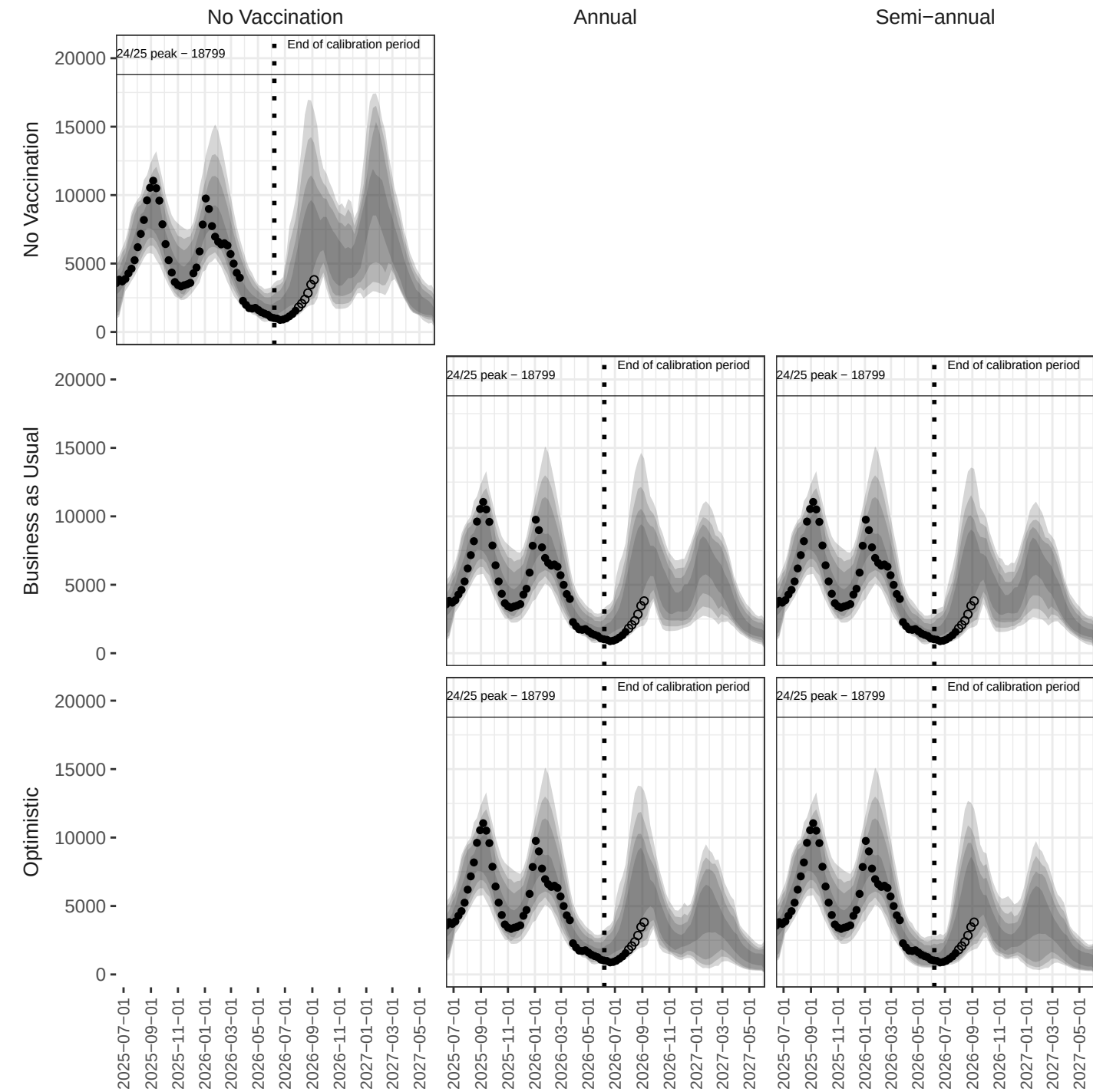
Percent and number of deaths averted by immunization



National Ensemble Projections

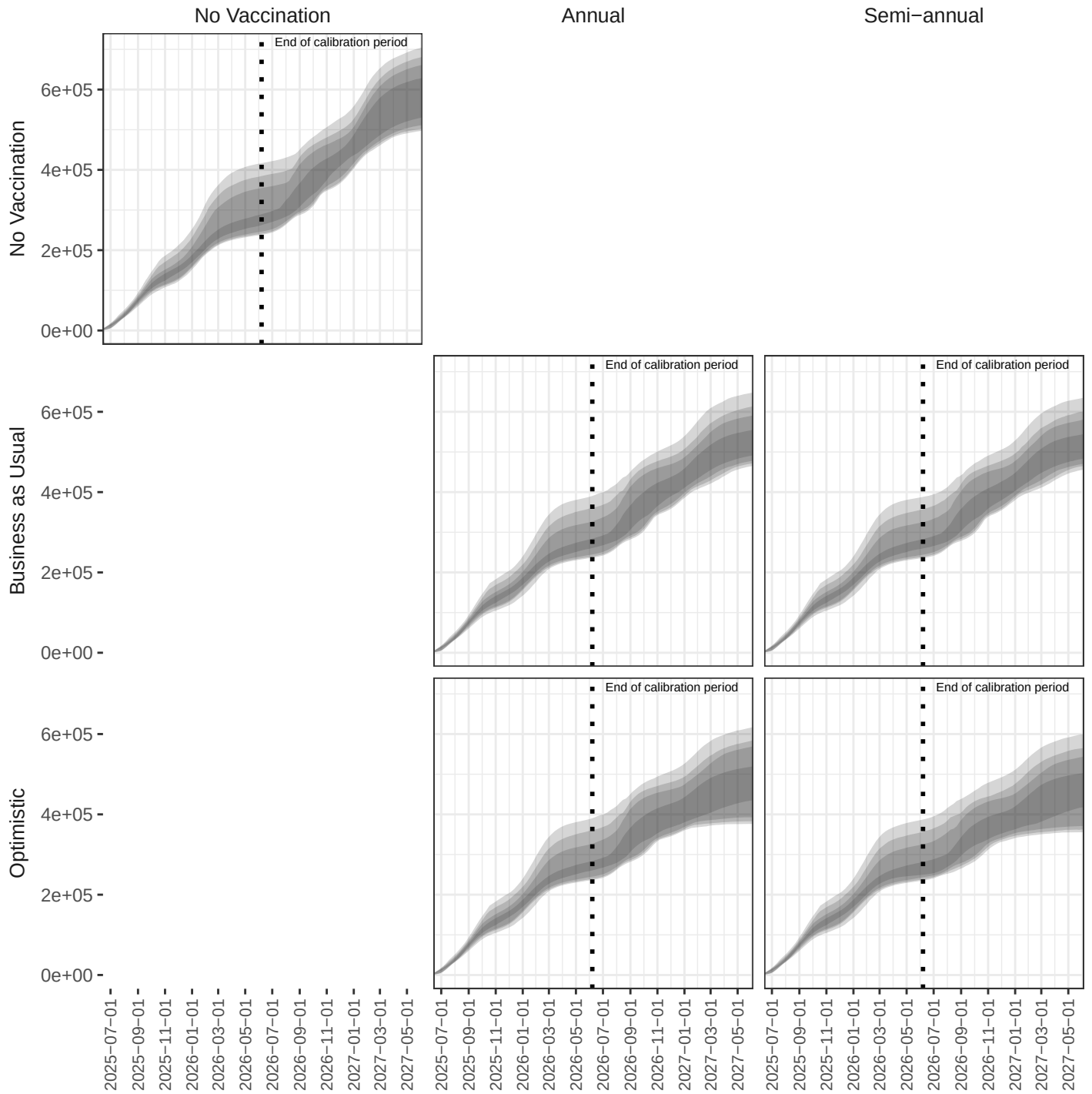
Weekly Incident hospitalizations in the national ensemble. Horizontal lines are given for prior peak incident hospitalizations, from past 2024-25 season based on NHSN data.

US ensemble projection intervals – Hospitalizations



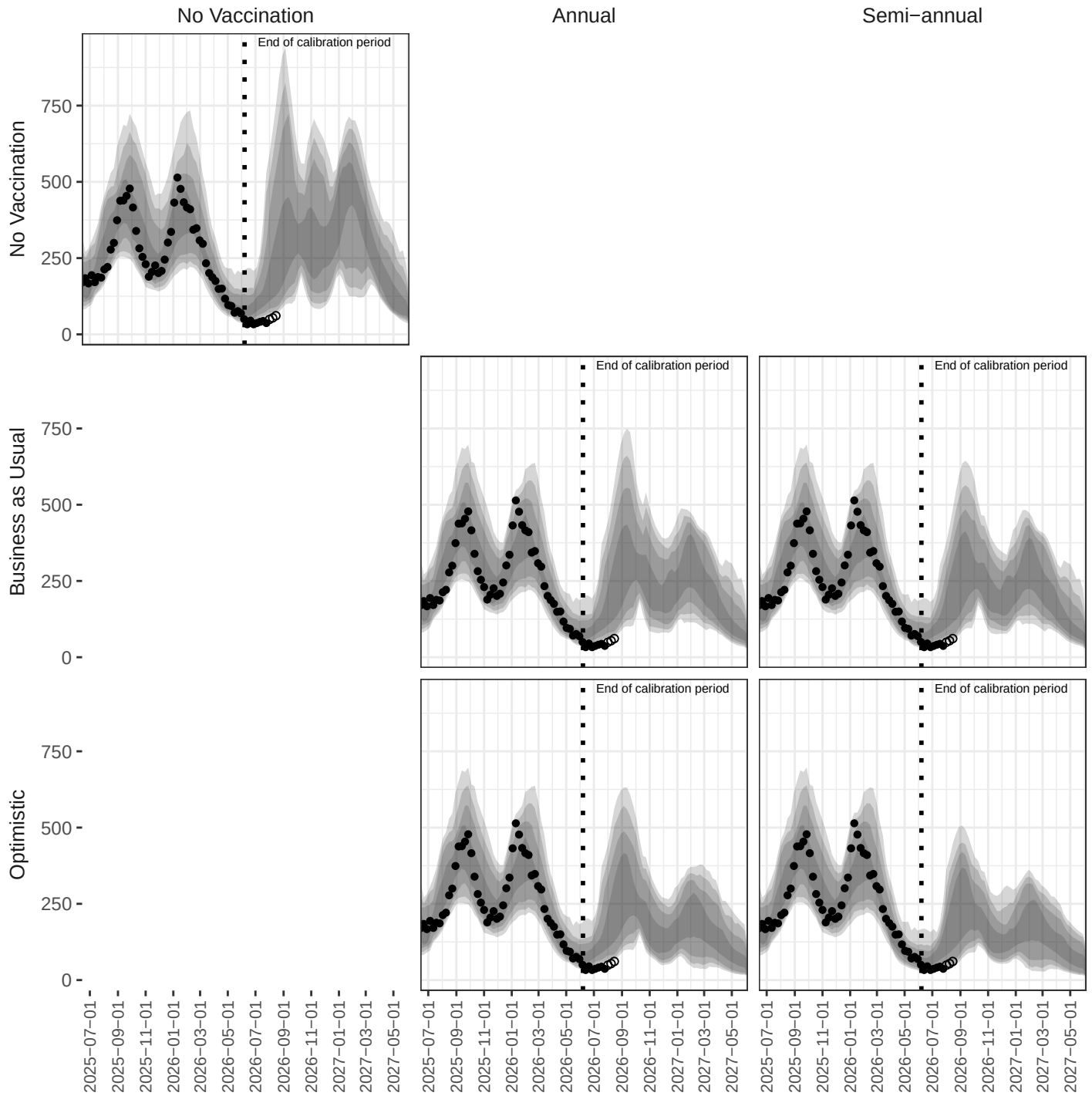
From lightest to darkest shading represents 95%, 90%, 80% and 50% projection intervals

US ensemble projection intervals – Cumulative Hospitalizations



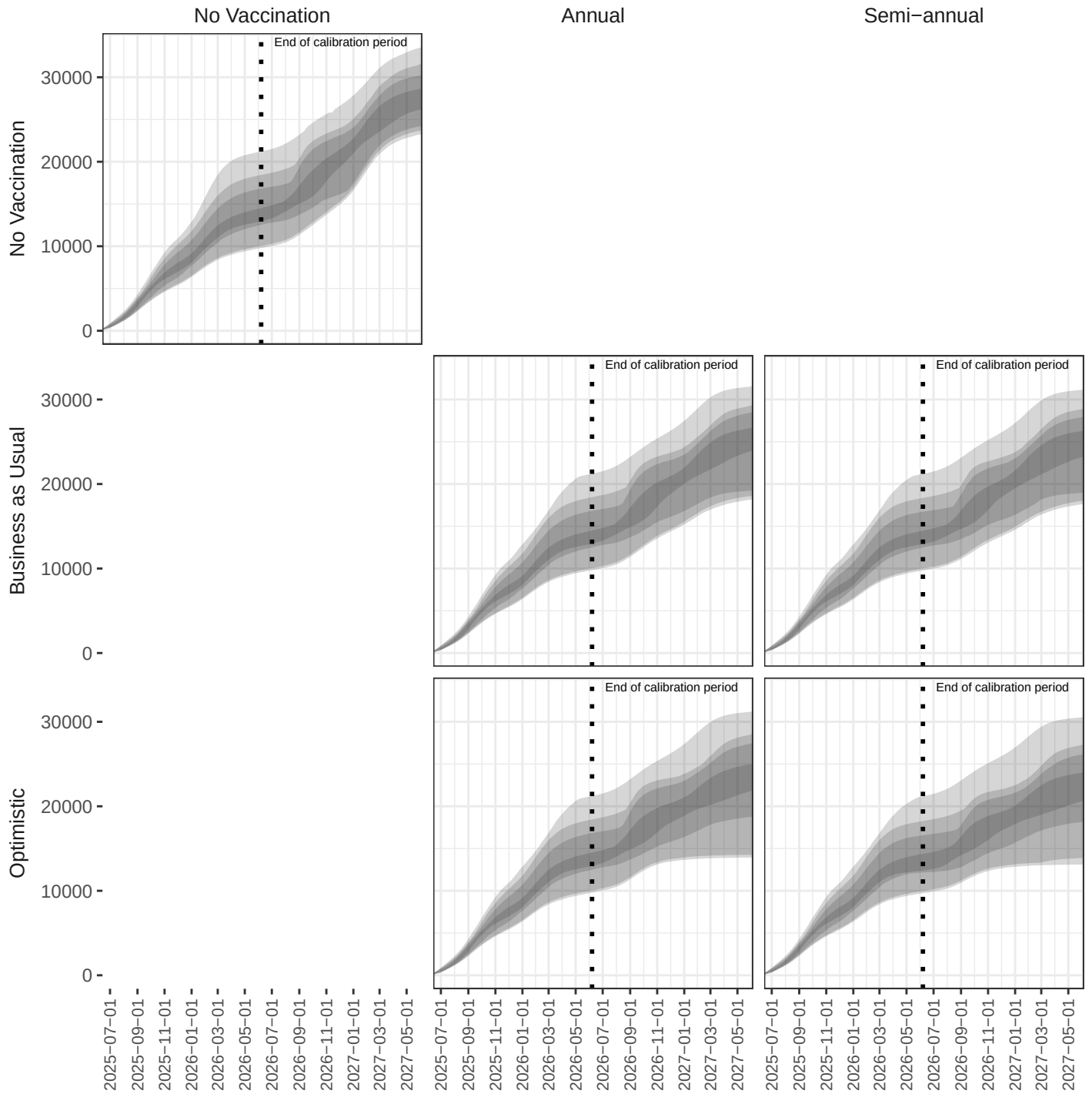
From lightest to darkest shading represents 95%, 90%, 80% and 50% projection intervals

US ensemble projection intervals – Deaths



From lightest to darkest shading represents 95%, 90%, 80% and 50% projection intervals

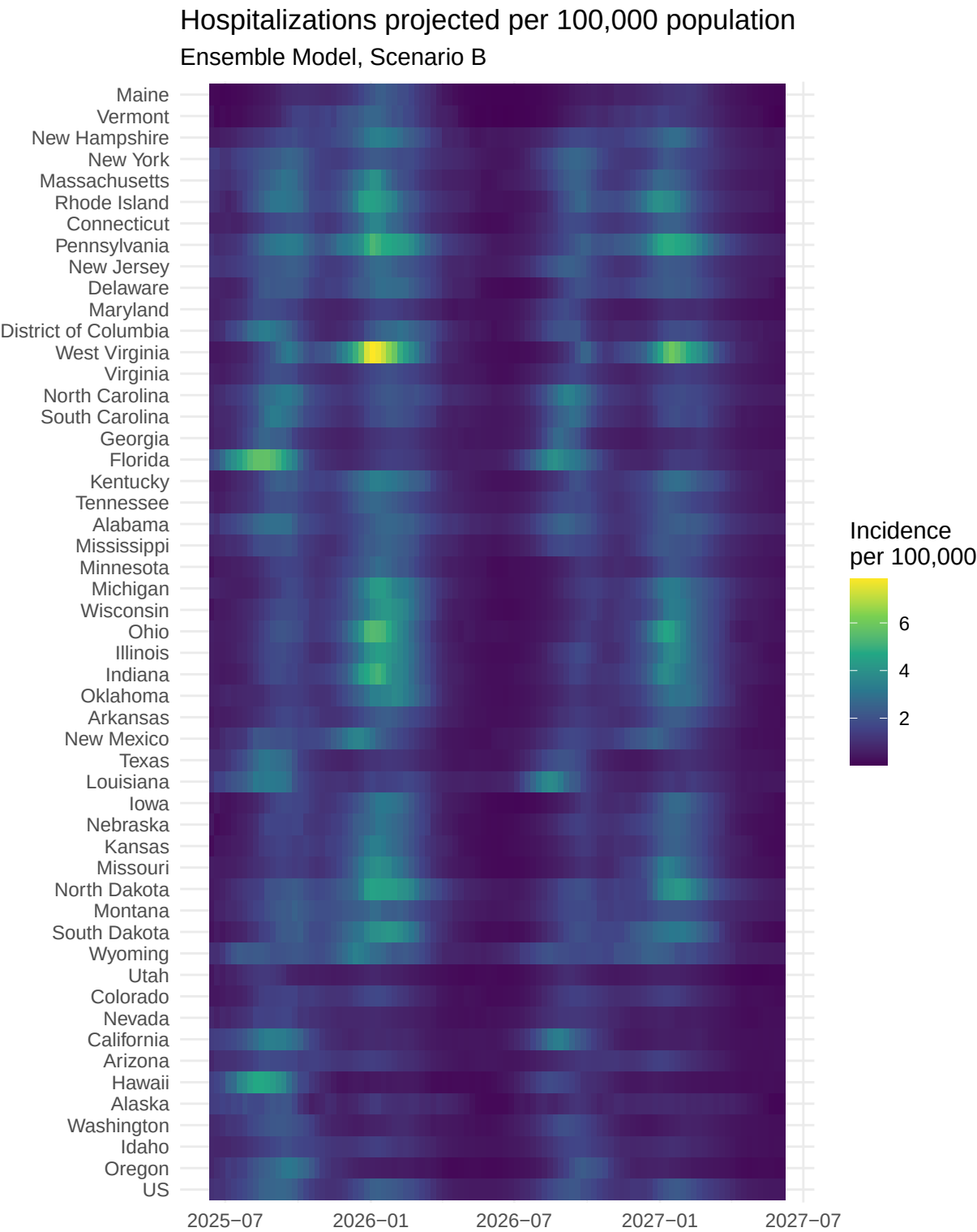
US ensemble projection intervals – Cumulative Deaths



From lightest to darkest shading represents 95%, 90%, 80% and 50% projection intervals

Spatiotemporal waves

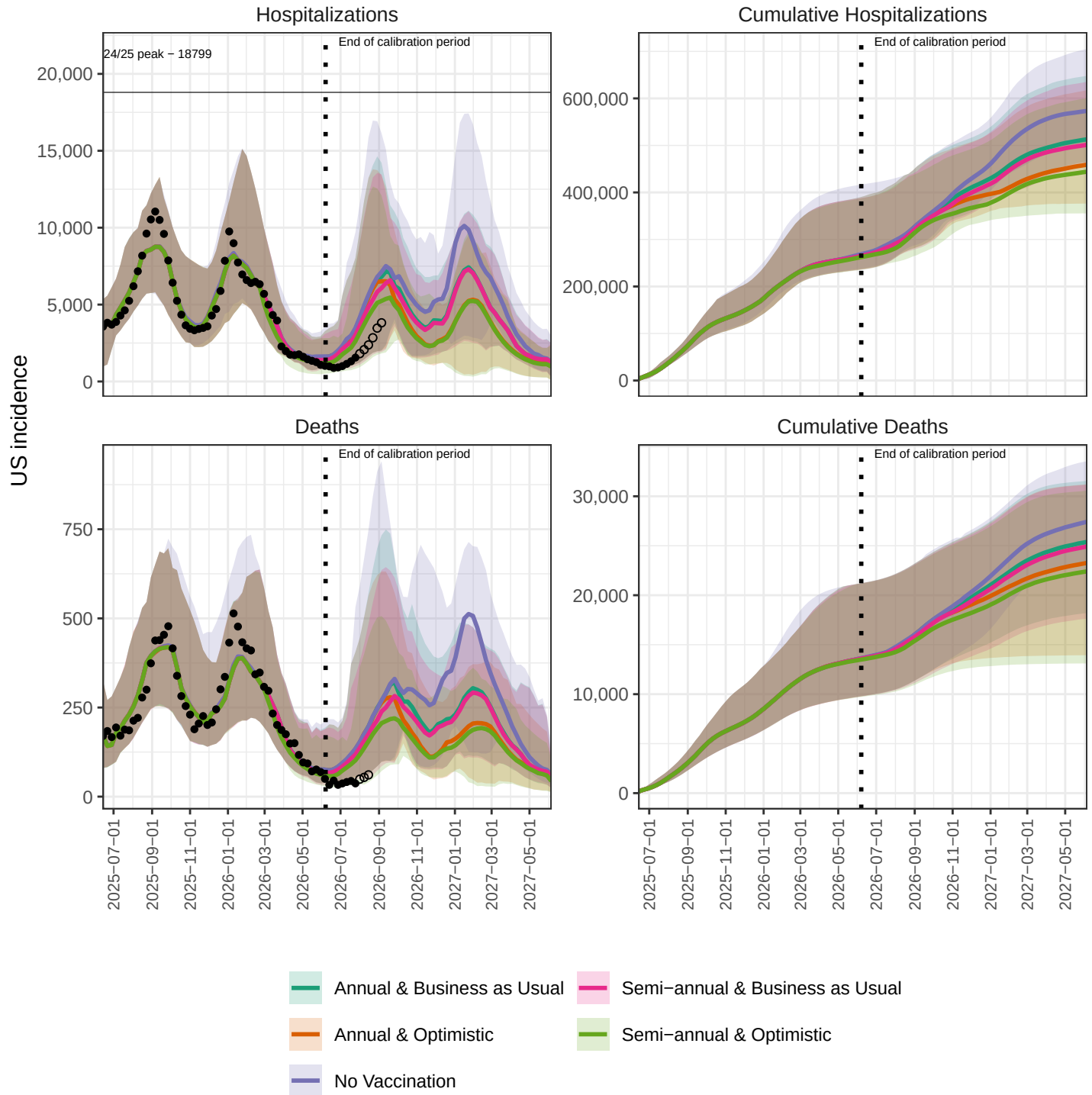
Median projected incident hospitalization per 100,00 population, by US State. (2022 population)



Ensemble Projection Comparisons

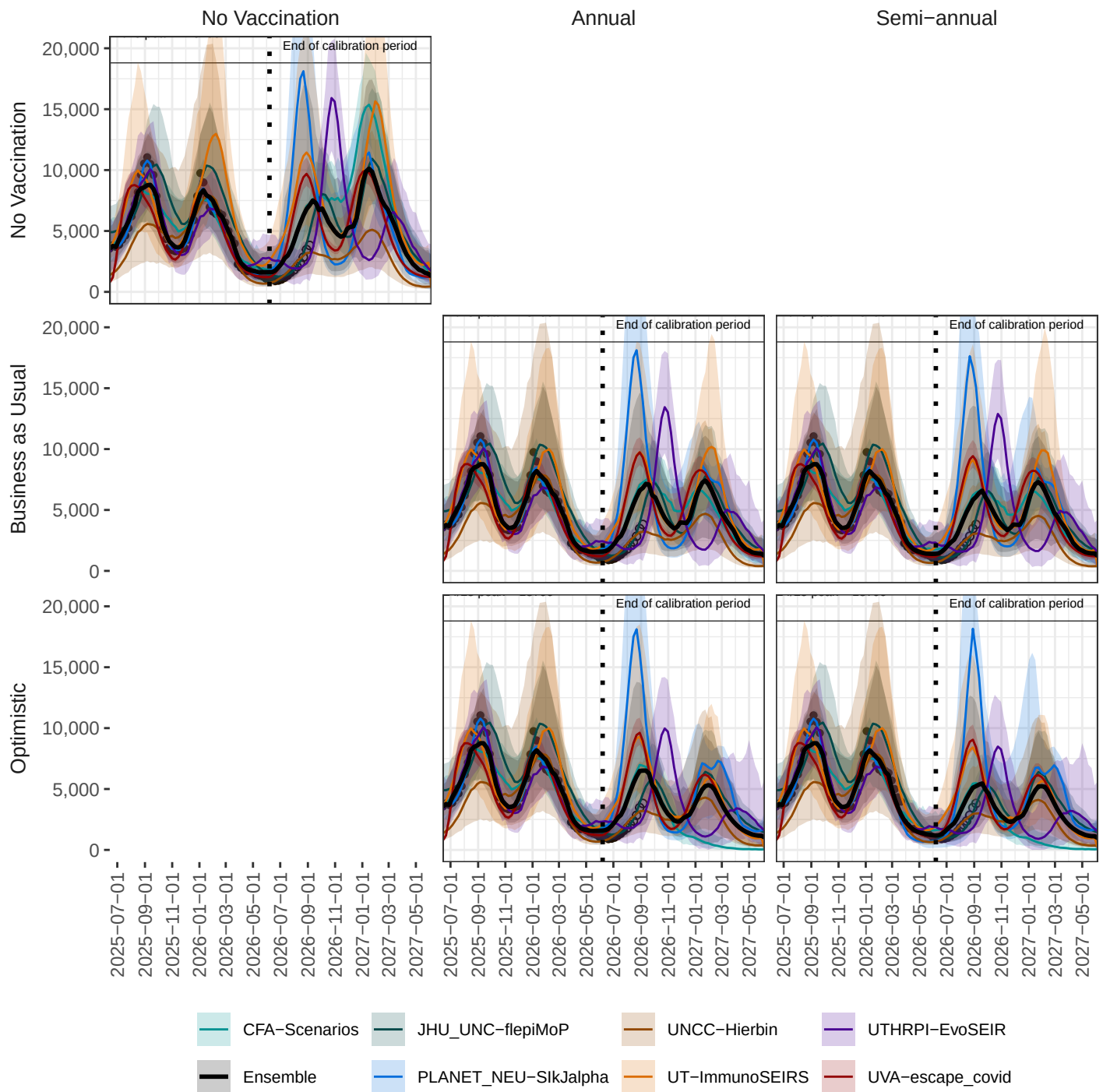
Ensemble projections for national incident and cumulative hospitalization and death separated by scenario

US ensemble projections & 95% projection intervals

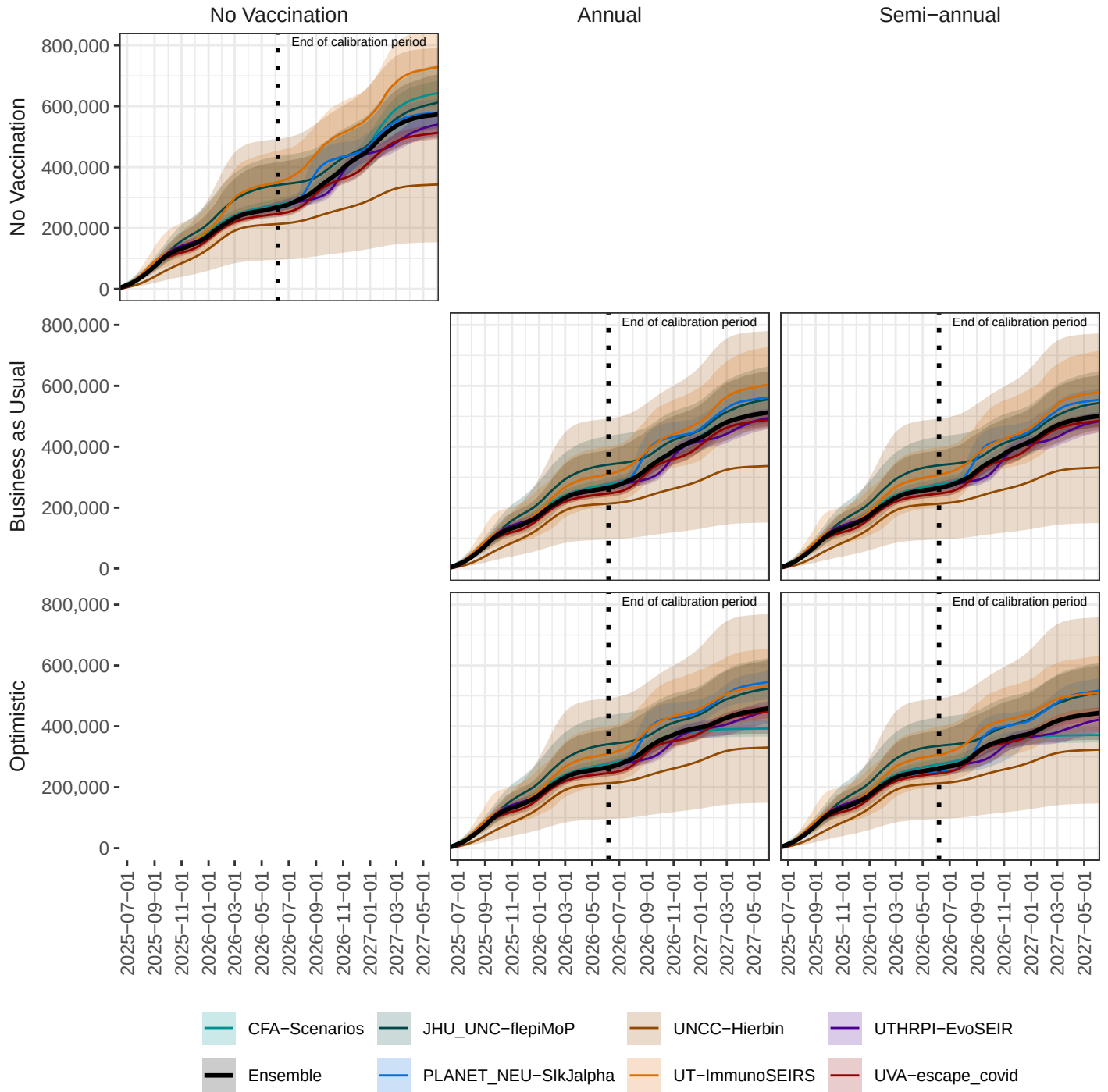


National individual model projections

US Individual Model Projections & 95% Projection Intervals Hospitalizations



US Individual Model Projections & 95% Projection Intervals Cumulative Hospitalizations



Teams and models

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- University of Texas Health Science Center at Houston - Rensselaer Polytechnic Institute - EvoSEIR
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